

structure, favourable scalability, potentially low cost, low power consumption, multiple state property, and three-dimensional stacking capability. However, the negative effects of toxic organic materials on the environment and human health pose great concerns for the application of these devices.

To overcome such problems, researchers and scientists have extensively investigated potential non-toxic biomaterials as substitutes over the last two decades. Biomaterials are biodegradable, bioresorbable, biocompatible, and environment friendly. More importantly, complicated chemical synthesis is not required to prepare natural biomaterials.

Previous reports have suggested several promising biomaterials to prepare organic-based electronic and optoelectronic devices with deoxyribonucleic acid that can serve as excellent electron-blocking layers in both green and blue organic light-emitting diodes.

Such materials offer promising uses for non-volatile transistor memory and write-once read-many-times memory applications. Researchers have also attempted to use different natural bio-

materials like tobacco mosaic virus, cysteine, ferritin protein, and enzymes to fabricate memory devices with various novel structures. These studies demonstrate that natural biomaterials can effectively simplify the fabrication process and reduce the manufacturing cost of organic memory devices.

A number of protein-based materials have successfully been utilised to fabricate resistive-type memories. For example, the cationic poly (allylamine hydrochloride) (PAH)/ferritin nanoparticle multilayer structure, which is prepared via layer-by-layer assembly, was recently demonstrated as a possible biomemory device with reliable switching property.

The electrochemical property of redox proteins was identified as the principal cause of reversible resistance-switching. The natural silk fibroin protein extracted from the cocoons of *Bombyx mori* (silkworm) has also been used as an active component of transparent bio-memristors. Gold (Au) nanoparticles embedded in silk fibroin protein have been used to improve the switching property.

However, most of the above-men-

tioned approaches involve complicated chemical processes, such as extraction or purification, which inevitably increase fabrication costs and method complexity.

To overcome many of such problems with the fabrication of biomaterials, researchers have come with the idea of fabricating and characterising non-volatile resistive switching memory devices by utilising chicken egg albumen obtained directly from fresh eggs without additional purification or extraction.

The proposed devices feature a simpler fabrication process as well as superior switching properties and reliability compared with most previous approaches.

Researchers have used cellulose as a dielectric layer in an inverter circuit and shellac as a dielectric in OFETs. Many other biological materials can be transformed into suitable dielectrics, including aloe, silk, and egg whites. Some other dielectrics with more exotic properties like hydrophobicity are also made from beeswax and carnauba wax.

Researchers are exploring to build OFETs in hard gelatin capsules or on caramelised sugar for biomedical appli-

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